$$\begin{aligned}
&\int \frac{x^2}{x^2-1} dx; x \in (-\infty; -1) \\
&= \int 1 + \left(\frac{1}{x^2-1}\right) dx \\
&= \int 1 dx + \int \frac{1}{x^2-1} dx \\
&= x + \int \frac{1}{x^2-1} dx
\end{aligned}$$

Dokažte

$$7 \mid 5^{2n+1} + 2^{2n+1} \ \forall n \in \mathbb{N}$$

zkusím indukcí?

$n = 1$:

$$7 \mid 5^3 + 2^3$$
$$7 \mid 125 + 8$$
$$7 \mid 133 \ \checkmark$$

$n + 1$:

$$7 \mid 5^{2n+3} + 2^{2n+3}$$
$$7 \mid (7k - 2^{2n+1}) \cdot 5^2 + 2^{2n+1} \cdot 2^2$$
$$7 \mid 7k \cdot 5^2 - 25 \cdot 2^{2n+1} + 4 \cdot 2^{2n+1}$$
$$7 \mid 7k \cdot 5^2 - 21 \cdot 2^{2n+1}$$
$$7 \mid 7k \cdot 5^2 + 7 \cdot (-3) \cdot 2^{2n+1} \ \checkmark$$

spočítejte

$$\begin{aligned}
&\lim_{x\to 0}\frac{\sqrt{2x+1}-1}{x^2-x}\\
&=\lim_{x\to 0}\frac{\left(\sqrt{2x+1}-1\right)'}{\left(x^2-x\right)'}\\
&=\lim_{x\to 0}\frac{\sqrt{2x+1}'}{2x-1}\\
&=\lim_{x\to 0}\frac{(2x+1)^{\left(\frac{1}{2}\right)'}}{2x-1}\\
&=\lim_{x\to 0}\frac{\frac{1}{2}(2x+1)^{-\frac{1}{2}}\cdot 2}{2x-1}\\
&=\lim_{x\to 0}\frac{(2x+1)^{-\frac{1}{2}}}{2x-1}\\
&=-1
\end{aligned}$$

spočítejte

$$\left(\sqrt{1-\cos(2x)}\right)'$$

$$\begin{aligned}
1-\cos(2x)&\geq 0\\
\cos(2x)&\leq 1\\
x&\in\mathbb{R}\\
D\left(\sqrt{1-\cos(2x)}\right)&=\mathbb{R}\\
H\left(\sqrt{1-\cos(2x)}\right)&=\langle 0,\sqrt{2}\rangle\\
\left(\sqrt{1-\cos(2x)}\right)'&=\left((1-\cos(2x))^{\frac{1}{2}}\right)'\\
&=\frac{1}{2}(1-\cos(2x))^{-\frac{1}{2}}\cdot 2\sin 2x\\
&=(1-\cos(2x))^{-\frac{1}{2}}\cdot\sin 2x\\
&=\frac{\sin 2x}{\sqrt{1-\cos(2x)}}\\
&=\frac{\sin 2x}{\sqrt{2\sin^2 x}}\\
&=\frac{\sin 2x}{\sqrt{2\sin^2 x}}\\
&=\frac{2\sin x\cos x}{\sqrt{2\sin^2 x}}
\end{aligned}$$

jednostranné derivace

$$D\left(\frac{\sin 2x}{\sqrt{2\sin^2 x}}\right)=\mathbb{R}\setminus\{k\pi\mid k\in\mathbb{Z}\}$$

derivace zprava např. v $x=0$

$$
\begin{aligned}
&\lim_{x\to 0^+} \frac{\sin 2x}{\sqrt{2\sin^2 x}} \\
&= \lim_{x\to 0^+} \frac{2x}{\sqrt{2x^2}} \\
&= \lim_{x\to 0^+} \frac{2x}{x\sqrt{2}} \\
&= \frac{2}{\sqrt{2}} \\
&= \sqrt{2}
\end{aligned}
$$

derivace zleva např. v $x = 0$

$$
\begin{aligned}
&\lim_{x\to 0^-} \frac{\sin 2x}{\sqrt{2\sin^2 x}} \\
&= \lim_{x\to 0^+} \frac{2x}{\sqrt{2x^2}} \\
&= \lim_{x\to 0^+} \frac{2x}{|x|\ \sqrt{2}} \\
&= -\frac{2}{\sqrt{2}} \\
&= -\sqrt{2}
\end{aligned}
$$

$$\int \frac{e^x+1}{e^{2x}+2e^x-3}dx$$
$$= \int \frac{e^x+1}{(e^x+3)(e^x-1)}dx$$
$$= \int \frac{e^x+1}{(e^x+3)(e^x-1)}dx$$

asi parciální zlomky ne?

$$\frac{A}{e^x+3}+\frac{B}{e^x-1} = \frac{e^x+1}{(e^x+3)(e^x-1)}$$
$$\frac{A(e^x-1)+B(e^x+3)}{(e^x+3)(e^x-1)} = \frac{e^x+1}{(e^x+3)(e^x-1)}$$
$$A(e^x-1)+B(e^x+3) = e^x+1$$
$$Ae^x-A+Be^x+3B = e^x+1$$

$$A+B = 1$$
$$3B-A = 1$$
$$A = 3B-1$$
$$3B-1+B = 1$$
$$4B = 2$$
$$B = \frac{1}{2}$$
$$A = \frac{3}{2}-1$$
$$A = \frac{1}{2}$$

$$= \int \frac{1}{2(e^x+3)}+\frac{1}{2(e^x-1)}dx$$
$$= \int \frac{1}{2(e^x+3)}dx+\int \frac{1}{2(e^x-1)}dx$$
$$= \frac{1}{2}\int \frac{1}{e^x+3}dx+\frac{1}{2}\int \frac{1}{e^x-1}dx$$